Contents

1. Units & Standard Conversions
2. Introduction — Common Instrument Problems
3. Measuring Range versus Accuracy
4. High Range / Long Scale Displays
5. Ergonomy & Standard Panel Layouts
6. Analogue versus Digital Presentation
7. Electronic (Glass) Displays
8. Readability & Parallax
9. Coloured Arcs & Colour Standardization
10. Practice Questions & Detailed Answers
11. Master Reference Tables

GHOST AVIATOR

GHOST AVIATOR

Join live classes for guided practice and doubt clearing.

1. Units & Standard Conversions

> **What this section covers:** the measurement units used throughout instrumentation, and the exact conversion values you are expected to know verbatim for the DGCA exam.

Pilots must move fluently between metric, imperial and aviation-specific units. The values below are quoted exactly from the source and must be memorised — examiners test the precise figures.

Quantity	Units	Key conversions (verbatim)
Distance	Metres, kilometres, feet, nautical & statute miles	1 NM = **6080 ft** or **1852 m**; 1 statute mile = **5280 ft** or **1609 m**
Speed	Knots, mph, km/h	1 knot = 1 NM per hour
Mass	Kilogram, pound	—
Pressure	hPa/mb, inHg, mmHg, psi	1 Atmos = **1013.25 hPa** = **1013.25 mb** = **29.92 inHg** = **760 mmHg** = **14.7 psi**
Temperature	Celsius, Kelvin (Absolute), Fahrenheit	MSL = **+15°C** = **+288 K** = **+59°F**
Volume	Litres, pints, gallons (Imp & US)	1 Imp gallon = **1.2 US gallon**
Angles / Position	Degrees, minutes, seconds	Latitude & longitude shown in Deg Min Sec

> **Exam Tip:** The two most-tested values are **1 NM = 6080 ft** (note: the ICAO/SI nautical mile is 1852 m) and the standard atmosphere set **1013.25 hPa / 29.92 inHg / +15°C**. Learn the whole standard-atmosphere line as one block.

2. Introduction — Common Instrument Problems

> **What this section covers:** why all instrumentation — from a simple dial to a glass cockpit — shares the same four design challenges.

Pilots receive information about the aircraft's state — speed, altitude, position and attitude — through instruments and displays. Regardless of vintage, every instrumentation system must contend with four general

Join live classes for guided practice and doubt clearing.

characteristics:

- **Range** — the span of values it can display.
- **Resolution** — the smallest change it can show.
- **Accuracy** — how closely the indication matches truth.
- **Reliability** — consistency of correct indication over time.

3. Measuring Range versus Accuracy

> **What this section covers:** the fundamental design trade-off between showing a wide range and reading it accurately, and how scale shape (linear vs non-linear) is used to manage it.

It is often necessary to show a large operating range yet still indicate accurately across the whole range. For example, an airliner limited to a maximum permitted airspeed of **350 knots** might use an instrument designed to display up to **380** or **400 knots**. But safety-critical speeds must be read to the nearest knot. If the entire range is squeezed onto a single revolution, each one-knot division becomes too small to read accurately — that is the range-versus-accuracy conflict.

Circular scale — linear

A simple indicator showing values over a range of 0 to 30 units uses evenly spaced graduations; the required accuracy governs the spacing.

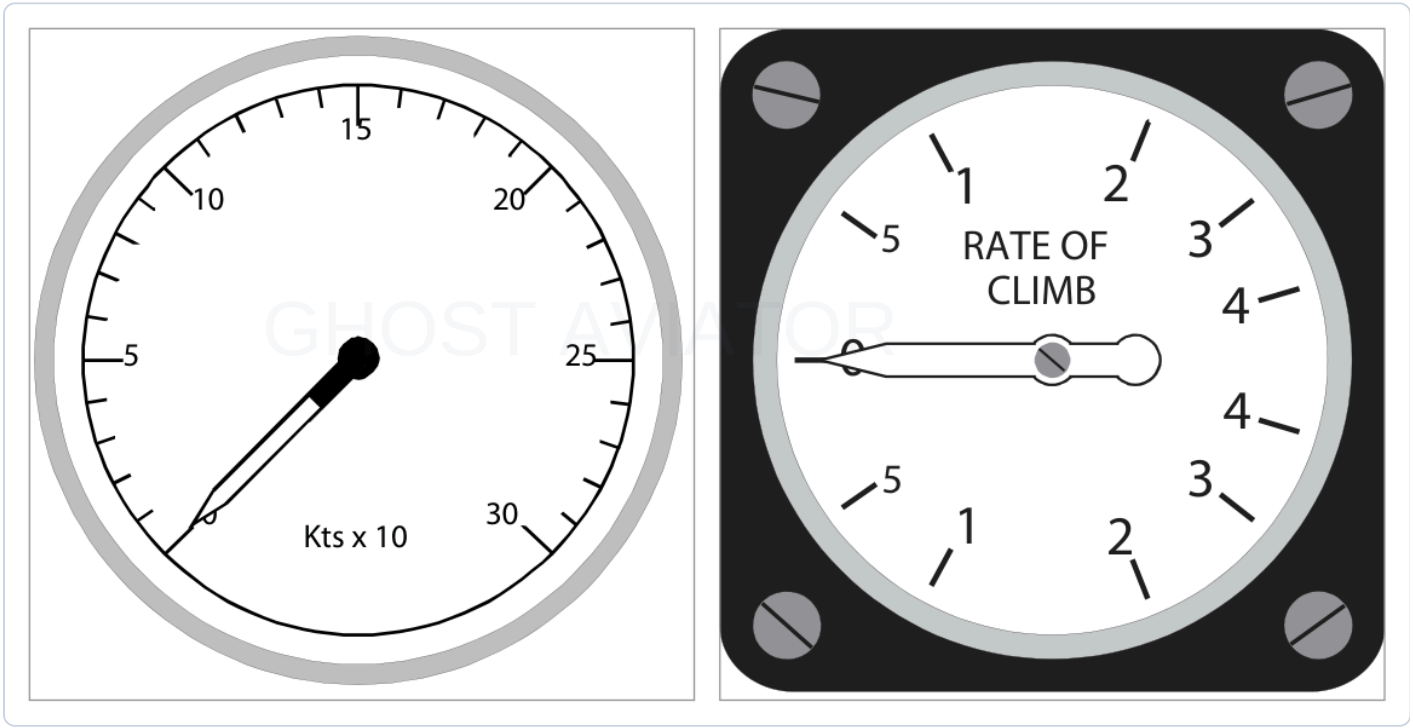


Fig 1.1 / 1.2 — Circular scale, linear (left) and non-linear/logarithmic (right) —

Circular scale — non-linear

Join live classes for guided practice and doubt clearing.

Some instruments must show changes more accurately at certain parts of the scale. A rate-of-climb indicator uses a **logarithmic scale** so that low rates of climb/descent are spread out and easily read, while high rates are compressed.

4. High Range / Long Scale Displays

> **What this section covers:** four design solutions used when 360° of pointer travel cannot cover the required range with adequate accuracy.

Solution	How it works	Example reading
Single pointer, multiple revolutions	Pointer makes >1 revolution to cover range — can cause confusion	ASI showing **300 kt** (Fig 1.3)
Moving pointer + moving scale	Pointer over fixed scale (tens), inset moving scale shows hundreds	ASI showing **33 kt** (Fig 1.4)
Two concentric pointers	Small needle/inner scale reads tens; large needle/outer scale reads units	Rev counter showing **25½ % rpm** (Fig 1.5)
Clock-style (three pointer)	Like hours/minutes/seconds — used on many altimeters	Altimeter showing **25 950 ft** (Fig 1.6)



Fig 1.4 / 1.5 — Inset moving-scale indicator / revolution counter —

Join live classes for guided practice and doubt clearing.

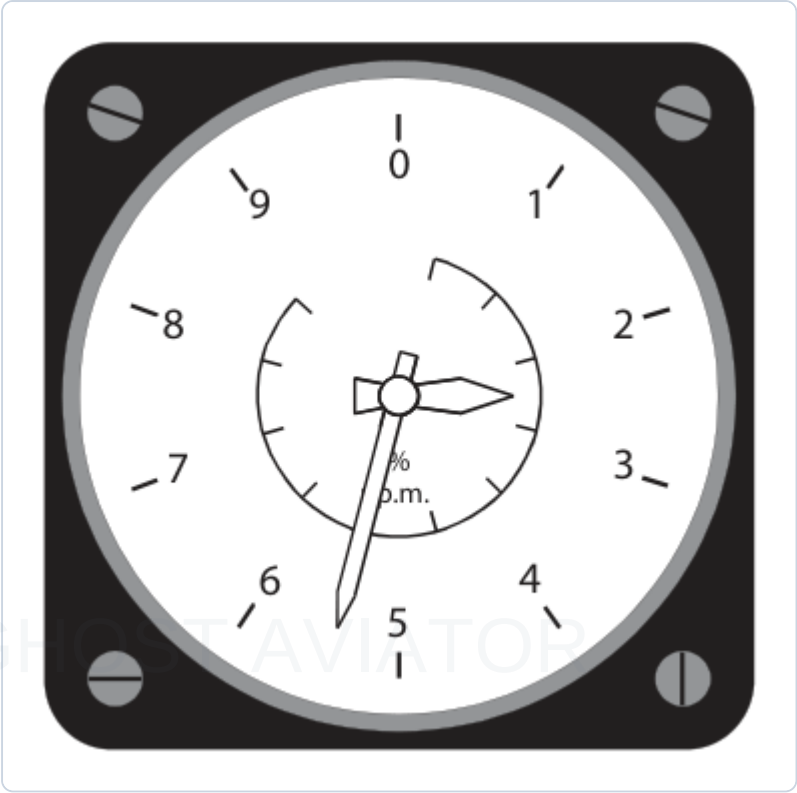


Fig 1.5 / 1.6 — Concentric / three-pointer style display —

Exam Tip — three-pointer altimeter: long pointer = **100 ft** per division (1000 ft/rev); middle pointer = **1000 ft** per division (10 000 ft/rev); smallest pointer = **10 000 ft** per division (100 000 ft/rev). The three-pointer altimeter has historically caused misreading accidents — this is why digital/counter displays were introduced.

5. Ergonomy & Standard Panel Layouts

What this section covers: human engineering of instruments, and the evolution from the "basic six" to the "basic T" layout.

Ergonomy (human engineering) is the science of the relationship between people and machines. For instruments it means designing displays unlikely to be misread and arranging them so interpretation is easy and correct.

From the "basic six" to the "basic T"

The flying instruments were first arranged as the **basic six**; other instruments were scattered to suit the manufacturer. Developments then led to the standard **basic T**.

Join live classes for guided practice and doubt clearing.

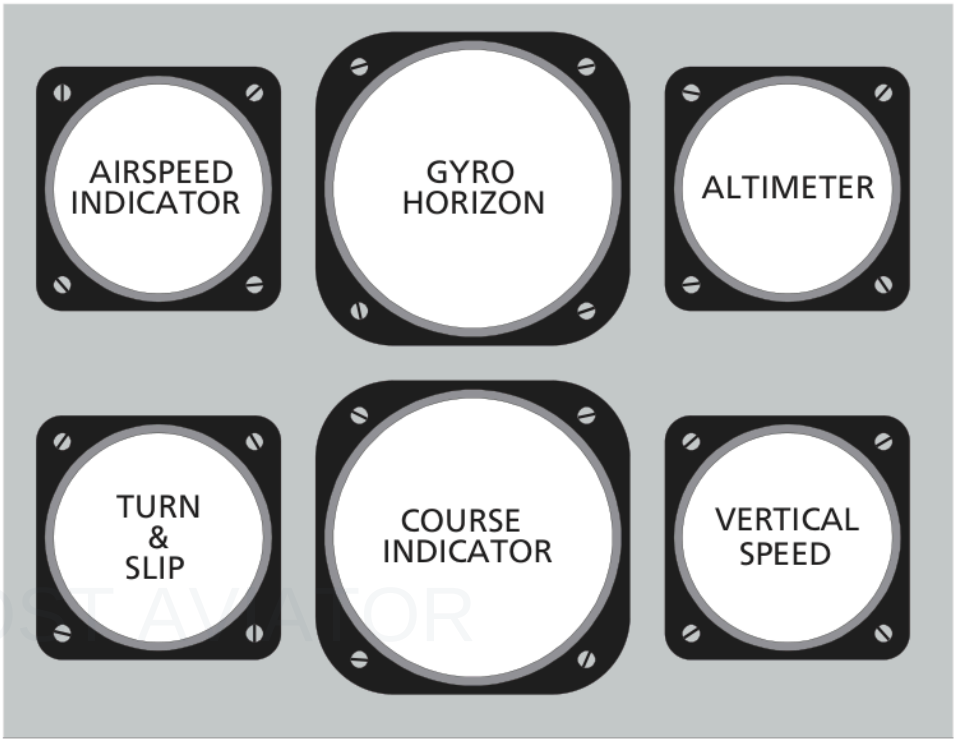
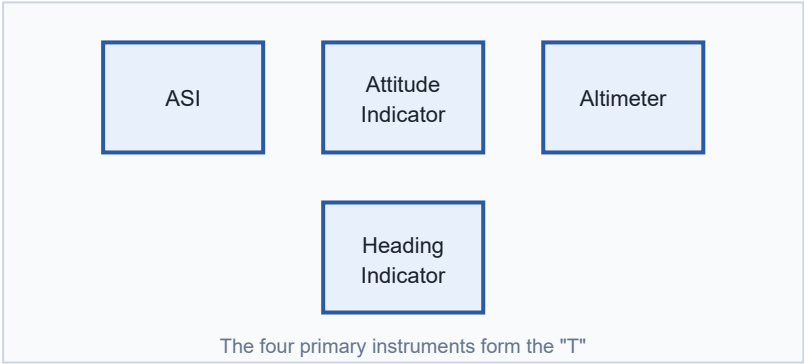


Fig 1.8 / 1.9 — The "basic T" layout / PA-34 Seneca basic panel —



The four primary instruments form the "T"

6. Analogue versus Digital Presentation

> **What this section covers:** the strengths of each presentation type, and why even "digital" altimeters retain a pointer.

Presentation can be **analogue** (a pointer on a dial) or **digital** (a row of numbers). With a 3-pointer analogue altimeter, an altitude such as **24 020 ft** is harder to absorb at a glance than a digital readout. Digital numbers are easier to read accurately.

Join live classes for guided practice and doubt clearing.

Why a pointer survives on digital displays: the human eye and brain cannot easily interpret *rate* information from moving numbers. Pilots pick up secondary rate-of-change information from the angular rate of a moving pointer, so a pointer is retained even on mainly-digital altimeters.

7. Electronic (Glass) Displays

What this section covers: the architecture of modern electronic flight decks and where the computing hardware physically lives.

Traditionally instruments lived on the instrument panel. With modern electronic displays, the displays remain on the flight deck where the crew can see and operate them, but the computing and power units are located remotely — usually in a separate compartment called the **Avionics Bay** or the **Electrics and Electronics (E&E) Bay**.

GHOST AVIATOR

```
flowchart LR
S[Sensors / Probes] --> C[Computing units
in Avionics / E&E Bay]
P[Power units] --> C
C --> D[Flight-deck displays
CRT / LCD glass screens]
D --> Crew([Flight Crew])
```

8. Readability & Parallax

What this section covers: the eye reference point and the parallax error that arises when it is not respected.

GHOST AVIATOR

A readable instrument is designed around an **eye reference point** — the anticipated position of the pilot's eye in normal viewing. Where an index/reference mark sits in front of a scale, the eye, index and scale must all be in line.

Parallax error: viewing an instrument from slightly to one side instead of from the front causes the index to appear against the wrong part of the scale. This is a reading error, not an instrument fault, and is avoided by viewing from the design eye reference point.

9. Coloured Arcs & Colour Standardization

Join live classes for guided practice and doubt clearing.

What this section covers: the standard colour codes for conventional instruments and the extended CS-25 colour set for electronic displays.

Conventional (non-electronic) instruments

Colour	Meaning
Green	Normal operating range
Yellow / Amber	Cautionary range
Red	Warning, or unsafe operating range

Electronic displays — CS-25 standardization

GHOST AVIATOR

Colour	Meaning
White	Present status
Blue	Temporary situation
Green	Normal operating range
Yellow / Amber	Cautionary range
Red	Warning, or unsafe operating range

Quick Revision Summary: Four instrument problems = range, resolution, accuracy, reliability. Standard atmosphere = **1013.25 hPa / 29.92 inHg / 760 mmHg / 14.7 psi / +15°C / +288 K / +59°F**. 1 NM = **6080 ft**. Three-pointer altimeter: 100 / 1000 / 10 000 ft per division. Pointer kept on digital displays for *rate* sensing. Parallax = off-axis viewing error. Conventional colours: green/amber/red; CS-25 adds white & blue.

GHOST AVIATOR

Practice Questions & Detailed Answers

Chapter 1 of the source is descriptive and contains no question bank, so the questions below are **instructor-generated** in the DGCA style, tied directly to the section content above. Attempt each before revealing the answer.

Q1. In the ICAO standard atmosphere at mean sea level, the pressure is:

a. 1000 hPa / 28.00 inHg

b. 1013.25 hPa / 29.92 inHg

c. 1025 hPa / 30.12 inHg

d. 1013.25 inHg / 760 hPa

Correct Answer: **(b)** 1013.25 hPa / 29.92 inHg

Explanation: The standard MSL pressure set is **1013.25 hPa = 29.92 inHg = 760 mmHg = 14.7 psi**. See Section 1 above.

Why the other options are wrong:

- **(a)** — 1000 hPa / 28.00 inHg are not the standard values.
- **(c)** — high-pressure values, not standard.
- **(d)** — units transposed (inHg and hPa swapped).

Instructor's Note: *Memorise the full standard-atmosphere line as a single block; it recurs in altimetry and air-data questions.*

Q2. One nautical mile is equal to:

a. 5280 ft

b. 1609 m

c. 6080 ft

d. 1000 m

Correct Answer: **(c)** 6080 ft (also 1852 m)

Join live classes for guided practice and doubt clearing.

Explanation: Per Section 1, 1 NM = **6080 ft** or **1852 m**.

Why the other options are wrong:

- **(a)** & **(b)** — 5280 ft / 1609 m define the *statute* mile.
- **(d)** — 1000 m is one kilometre.

Instructor's Note: *Don't confuse the nautical mile (6080 ft) with the statute mile (5280 ft).*

Q3. A pointer is retained on an otherwise digital altimeter primarily because:

a. it is cheaper to manufacture

b. the eye cannot easily read rate information from moving numbers

c. regulations forbid fully digital altimeters

d. it improves the instrument's accuracy

Correct Answer: **(b)** the eye cannot easily read rate information from moving numbers

Explanation: Section 6 — pilots pick up rate-of-change information from the angular rate of a moving pointer, which digits cannot convey at a glance.

Why the other options are wrong:

- **(a)** — cost is not the stated reason.
- **(c)** — no such prohibition.
- **(d)** — the pointer aids rate sensing, not accuracy of the value.

Instructor's Note: *Same logic underlies the "trend vector" on modern tape displays.*

Q4. Viewing an instrument from slightly off to one side produces:

Join live classes for guided practice and doubt clearing.

a. position error

b. parallax error

c. hysteresis

d. lag error

Correct Answer: **(b)** parallax error

Explanation: Section 8 — parallax is the reading error from not viewing along the eye reference point/index/scale line.

Why the other options are wrong:

- **(a)** — position error is a pitot/static pressure error (Ch.2).
- **(c)** & **(d)** — hysteresis and lag are mechanical/dynamic errors, not viewing-angle errors.

Instructor's Note: *Parallax is a reading error — the instrument itself is correct.*

Q5. On a conventional instrument, an amber (yellow) arc indicates:

a. normal operating range

b. cautionary range

c. warning / unsafe range

d. present status

Correct Answer: **(b)** cautionary range

Explanation: Section 9 — green = normal, yellow/amber = caution, red = warning/unsafe.

Why the other options are wrong:

- **(a)** — green.
- **(c)** — red.

Join live classes for guided practice and doubt clearing.

- **(d)** — "present status" is white, and only in the CS-25 electronic set.

> ***Instructor's Note:*** *White and blue appear only in the CS-25 electronic-display colour set.*

Master Reference Tables

All numerical values in this chapter

Value	Meaning	Section
1 NM = 6080 ft = 1852 m	Nautical mile	1
1 statute mile = 5280 ft = 1609 m	Statute mile	1
1 Imp gal = 1.2 US gal	Volume conversion	1
1013.25 hPa / mb	Std MSL pressure	1
29.92 inHg / 760 mmHg / 14.7 psi	Std MSL pressure (other units)	1
+15°C / +288 K / +59°F	Std MSL temperature	1
350 / 380–400 kt	Example Vmax vs scale design	3
100 / 1000 / 10 000 ft per division	Three-pointer altimeter	4

Memory aids

> "**4 R's**" of instrument problems: **R**ange, **R**esolution, accu**R**acy, **R**eliability. Colour order (conventional): **G**reen–**A**mber–**R**ed = Go / Attention / Risk.

Answer key

Q1	Q2	Q3	Q4	Q5
b	c	b	b	b

Join live classes for guided practice and doubt clearing.